

MINI PROJECT

MotoMonitor – Smart Vehicle Service and Maintenance Management System

GROUP NO-3

Submitted by

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Abstract

MotoMonitor – Smart Vehicle Service and Maintenance Management System

MotoMonitor is a web-based platform designed to help vehicle owners stay on top of important service and document deadlines. Many owners miss renewal dates for insurance, road tax, pollution certificates, driving licenses, and routine servicing, leading to legal penalties, unexpected breakdowns, increased repair costs, and safety risks. MotoMonitor addresses this by tracking these dates and sending timely, escalating reminders — starting with an email well in advance, followed by SMS as the deadline nears, along with a weekly summary digest listing everything due across all of a user's vehicles.

Beyond reminders, the system acts as a personal record keeper. Users can upload and securely store documents such as insurance papers, RC copies, and pollution certificates, log completed services along with their cost and notes, and manage multiple vehicles under a single account. A simple dashboard displays upcoming renewals and spending trends, while a dedicated overdue tracker highlights lapsed items that need urgent attention, helping users stay compliant and avoid unnecessary risk.

In addition, the platform introduces a bulk-purchase marketplace for bike accessories and spare parts. By aggregating demand from multiple users for the same product, the platform can source items at wholesale rates and pass on the savings, allowing users to purchase accessories at prices lower than standard retail. This ties naturally into the maintenance tracking feature, as users can be recommended and directly purchase the parts they are due to replace.

The system is developed using HTML5, CSS3, and JavaScript for the frontend, Python for backend development, and MySQL as the database management system. Email notifications through SMTP and SMS services are integrated to ensure users receive timely alerts across multiple communication channels. By combining preventive maintenance tracking, digital record-keeping, and a cost-saving marketplace, MotoMonitor aims to reduce maintenance costs, improve road safety, and provide a reliable, comprehensive platform for vehicle service and maintenance management.

Technologies to be Used

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Python

Database

- MySQL

Notifications

- Email Notifications
- SMS

Tools

- Visual Studio Code
- Git & GitHub